

Darin Lei

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EDUCATION

B. Sc. Hons., Psychology Research Specialist (Thesis Program), Cognitive Science Major 2023—2027

University of Toronto

Honours Thesis in progress: Determining the mechanisms of autobiographical memory with chronic ambulatory intracranial recording

Advisor: Dr. Morgan D. Barense

cGPA: 3.90/4.00

ACADEMIC HONOURS

University of Toronto Excellence Award for Health Sciences (CA\$7,500) 2026

Dean's List Scholar of the Faculty of Arts & Science 2025—2026

Ontario Scholar Award 2023

DSC International School Award for Excellence in Language (HK\$2,000) 2023

DSC International School Honour Roll 2023

STUDENT-REVIEWED PUBLICATIONS

Lei, D., & Djedovic, A. (2025). The Synergistic Mind-Body (SAIS) Hypothesis: A Daoist Approach to Consciousness from the Perspective of Traditional Chinese Medicine. *Canadian Undergraduate Journal of Cognitive Science*, 8, e7230. <https://doi.org/10.21810/cujcs.v8i1.7230>

Lei, D., & Vervaeke, J. (2026). From 'Ha Ha!' to 'Aha!': Towards an Integrated Understanding of the Effect of Humour Comprehension on Insight Problem-Solving. *Mentis: Journal of Cognitive Science and Artificial Intelligence*, 1(1), 40–56.

PRESENTATIONS

Lei, D., Brachaniec, M., Hong, B., Hughes, E. C., & Barense, M. D. (2026, June 2). *The HippoCamera Knowledge Hub Program* [Conference presentation]. AGE-WELL Early Professionals, Inspired Careers in AgeTech 7th Annual Conference (EPIC-AT), Toronto, ON, Canada.

Lei, D., Hughes, E. C., Hong, B., & Barense, M. D. (2026, February 5–6). *The HippoCamera Knowledge Hub: Developing A Digital Educational Program For Older Adults To Promote Healthy Aging* [Poster presentation]. Lake Ontario Visionary Establishment 53rd Annual Conference (LOVE), The Brock Niagara Falls Fallview, Niagara Falls, ON, Canada.

Hong, B., **Lei, D.,** Hughes, E. C., & Barense, M. D. (2025, October 22–23). *Developing a digital educational program for older adults to improve memory: a person-based approach* [Conference presentation]. AGE-WELL 10th Annual Conference (AGEWELL2025), Palais des Congrès, Montréal, QB, Canada.

Lei, D., Hong, B., Hughes, E. C., & Barense, M. D. (2025, September 25). *The HippoCamera Educational Program: Developing a digital educational program to improve memory in older adults: a person-based approach* [Poster presentation]. University of Toronto St. George Campus Research Opportunity Program (ROP) Research Poster Fair, Toronto, ON, Canada.

Lei, D., & Vervaeke, J. (2025, April 8). *The "High" Mind: Towards a Unified Theoretical Account of Insight Problem-Solving, Predictive Processing, & Relevance Realization Through Oneirogenic Psychedelics* [Poster presentation]. University of Toronto St. George Campus Psychology Undergraduate Research Community (PURC) 2nd Annual Undergraduate Symposium, Toronto, ON, Canada.

RESEARCH EXPERIENCE

Multisensory Integration in Virtual Environments (MIVE) Lab, KITE Research Institute, Toronto Rehabilitation Institute, University Health Network

PI: Dr. Jennifer L. Campos

Toronto, ON

Individual Summer Research Project Student

May 2026—Present

- Conduct literature reviews on the relationship between hearing loss and vection, hearing loss and vestibular dysfunction, compensatory strategies for walking imbalances in auditory-impaired individuals, and sensorimotor integration with respect to balance.
- Attend weekly lab meetings on a wide range of topics including vection, hearing loss, technological innovations in geriatric care, and visually induced motion sickness (VIMS).
- Perform data analyses using electromyography (EMG) and posturography.
- Present ongoing work in my assigned project at lab meetings.
- Administer psychometric assessments including Dynamic Visual Acuity (DVA), Trail-Making Test, Stroop, Pure Tone Test, Canadian Digit Triplet Task (CDTT), Pelli-Robson Contrast Sensitivity Chart, Randot Stereo Test, MoCA, and Sound Induced Flash Illusion (SIFI), and a Rod and Disk Task to obtain baseline measures of cognitive, visual, auditory, and vestibular functions.
- **Main Project:** Visual perturbation in vection and hearing loss
 - It is well-established that individuals with hearing loss, especially older adults, tend to struggle with balance. To compensate for these deficits in balance, individuals may react either by stiffening their bodies or moving dynamically with the direction of motion. However, these compensatory strategies are diametrically opposed in what they suggest about the sensorimotor integration and vestibular function as a whole. This project seeks to determine not only what strategies are used in hearing loss to combat issues with balance but also the mechanisms that underlie such strategies, and their resultant effects on illusory perception of self-motion (vection).

Texas Computational Memory Lab, University of Texas Southwestern Medical Center

PI: Dr. Morgan D. Barense, Dr. Bradley Lega

Dallas, TX (Hybrid) & Toronto, ON

Visiting Scholar and Undergraduate Thesis Student

January 2026—Present

- **Main Project:** Determining the neural and cognitive mechanisms of temporal context recovery in ambulatory epilepsy patients
 - Episodic recollection depends on recovering not only the content of an event, but also its temporal context. Retrieved-context theories predict that recalling an episode reinstates its encoding context and thereby biases recovery toward temporally adjacent events. However, this key prediction has only been determined in highly controlled, list-based experiments which lack the rich personal context of autobiographical memory (AM). This project will leverage HippoCamera (HC) and the NeuroPace responsive neurostimulation (RNS) system together to make this problem tractable. This experiment will be the first to directly test retrieved-context mechanisms in real-world AM, establishing whether canonical models of temporal context govern memory as it unfolds in daily life.

Levine Lab, Rotman Research Institute, Baycrest Academy for Research and Education

PI: Dr. Brian Levine

Toronto, ON

Individual Research Project Student

January 2026—Present

- Conduct literature reviews on the memory functions of sleep (specifically with respect to the formation of episodic memories in autobiographical memory), targeted memory reactivation (TMR), and the use of naturalistic stimuli in memory research paradigms.
- Process behavioural datasets using multivariate statistical models to test hypotheses and predict outcomes.

- Analyse free recall interview data from participants using the Autobiographical Interview (Levine, 2002) and administer a battery of neuropsychological tests for determining baselines of key variables including the Montreal Cognitive Assessment (MoCA).
- Attend weekly lab meetings discussing ongoing research on human memory with a focus on individual differences in autobiographical memory and traumatic memories.
- Present ongoing work in my assigned project at lab meetings.
- **Main Project:** Investigating the role of sleep in episodic memory consolidation
 - Over a century of research from the cognitive and clinical neuroscientific literatures has consistently demonstrated an integral role of sleep in episodic memory consolidation and recollection. Although robust, much of the evidence supporting this finding comes from laboratory-based paradigms involving recognition-cued recall for unnaturalistic stimuli such as wordlists. As such, it remains unclear whether sleep benefits the free recall of episodic memories and what neurophysiological mechanisms might at play in such an effect. Here, we aim to examine the impact of sleep on free recall by using a naturalistic memory design in the form of a controlled hospital tour. This study is the first to investigate the role of sleep for free recall under the context of naturalistic memory.

Memory & Perception Lab, Department of Psychology, University of Toronto

PI: Dr. Morgan D. Barense

Toronto, ON

Human-Computer Interaction Research Assistant

May 2025—Present

- Conduct literature reviews on cognitive decline in older adults, behavioural change frameworks, and educational interventions for age-related memory decline.
- Develop an educational program (“Knowledge Hub”) or knowledge centre on the HippoCamera app where older adults can learn more about research-backed strategies to improve their memory through designing prototypes on Figma.
- Evaluate the usability and acceptability of the educational program by consulting older adults in the Greater Toronto Area in qualitative, semi-structured interviews.
- Perform quantitative data analyses through RStudio and qualitative data analyses (thematic interpretative analyses) using NVivo.
- Attend weekly lab meetings discussing ongoing research on human memory.
- Present ongoing work in my assigned projects at lab meetings.
- **Project 1:** Developing technology-based solutions to promote healthy aging
 - Aging is associated with physical and cognitive changes that can greatly affect the autonomy, freedom, and well-being of older adults. This project aims to explore the efficacy of technology-based solutions to address the unique needs of older adults through the development of an educational program on a digital memory augmentation app called HippoCamera. By implementing these educational modules, older adults can learn how to apply different memory strategies in their daily life with a powerful impact on memory retention.
- **Project 2:** Determining the neural and cognitive mechanisms of temporal context recovery in ambulatory epilepsy patients
 - Episodic recollection depends on recovering not only the content of an event, but also its temporal context. Retrieved-context theories predict that recalling an episode reinstates its encoding context and thereby biases recovery toward temporally adjacent events. However, this key prediction has only been determined in highly controlled, list-based experiments which lack the rich personal context of autobiographical memory (AM). This project leverages HippoCamera (HC) and the NeuroPace responsive neurostimulation (RNS) system together to make this problem tractable. This experiment will be the first to directly test

retrieved-context mechanisms in real-world AM, establishing whether canonical models of temporal context govern memory as it unfolds in daily life.

Visual Cognition Lab, Department of Psychology, University of Toronto

PI: Dr. Jay Pratt

Research Assistant

Toronto, ON

January 2025—May 2026

- Conducted literature reviews on visual attention, specifically event file binding and retrieval, and the role of affect in visual processing.
- Obtained informed consent and collect data in-person for several projects under the supervision of graduate students in the lab.
- Recruited and scheduled timeslots with participants on SONA and run experiments using PsychoPy.
- Performed data analyses using R Studio and JASP. Specifically used R Studio for plotting psychometric curves and JASP for data cleaning and running t-tests and ANOVAs.
- **Project 1: The Effect of Stimulus Duration on Temporal Order Perception**
 - In visual perception, individuals often perceive a stimulus as the first even if it is veridically the second. This project explores whether such an effect is a result of the stimuli duration in visual working memory or a bias in early visual processing.
- **Project 2: The Role of Action Features in Stimulus-Response (SR) Binding and Retrieval**
 - It has been suggested that all the features of a stimulus and actions are bound into a holistic representation (SR binding). This project explores the nuances of this binding procedure by testing whether and how a response with multiple features is integrated to and retrieved from such a representation.
- **Project 3: The Role of Affect in Stimulus-Response (SR) Binding and Retrieval**
 - Affective information influences both visual processing and response selection. This project investigates whether the affective features of objects are involved in the SR binding procedure and, if so, whether it is more closely associated with visual or action representation.

Language & Learning Lab, Dr. Eric Jackman Institute of Child Study, Department of Applied Psychology & Human Development, Ontario Institute for Studies in Education, University of Toronto

PI: Dr. Patricia A. Ganea

Research Assistant

Toronto, ON

April 2025—November 2025

- Conducted literature reviews on three areas of early cognition in children: counterfactual reasoning, hypothetical reasoning about complex and dynamic systems, and executive functioning and language acquisition as a mechanism for reasoning about possibilities.
- Recruited and scheduled participants using Lookit, an MIT-built research platform for running online experiments with infants, toddlers, and children.
- Contacted parents and daycares for recruitment on a weekly basis and update information as needed on the Developmental Research Database (DRDB) System.
- Obtained informed consent and demographic data for projects via REDCap.
- Regularly attended lab meetings to discuss papers of interest and give practice presentations for upcoming symposia and conferences.
- Designed surveys and questionnaires in online experiments using Qualtrics.
- **Project 1: Simulation Mindset**
 - It is known that the prior simulation of episodic events to decision-making can often improve motivation to change behaviour in adults. However, this has been rarely explored in children and its emergence in the human cognitive developmental trajectory is unknown. This project explores whether increased engagement or exposure to counterfactual reasoning facilitates behavioural change in children.
- **Project 2: Mutability**

- In daily life, we often think about possibilities of change for events we have experienced in the past. When thinking about what could possibly change in a past event, we think about the aspects of it that would have been the most easily modifiable (mutable) to afford a different outcome. This appears to be true in adults, but whether it is true in children is not unknown. In this project, we ask whether there are age-related differences in thinking about possibilities of change, and if so, what mechanisms may be responsible for such differences.

Cognitive Science & Artificial Intelligence Students' Association (CASA)

Toronto, ON

Publication Author

March—August 2024

- Authored a critical theoretical review paper that proposed humour has a facilitating effect on insight problem-solving and consolidated existing views of insight problem-solving as a dynamically coordinated system of cognitive flexibility.

RELEVANT EXPERIENCE

University of Toronto Karate Club, Club Sports, Sport & Rec, Faculty of Kinesiology & Physical

Education (KPE), University of Toronto

Toronto, ON

Club President

May 2026 — Present

- Attend biweekly meetings on club outreach and instructional and seminar-based lesson design.
- Lead various social activities within the club and managing the team's online presence.
- Liaison with University of Toronto Faculty of Kinesiology and Physical Education staff to arrange facilities access and training spaces for athletes in the club.
- Manage administrative tasks including organizing fees, training, grading, attendance and relations with external institutions such as the Ontario Karate Federation and other clubs across Ontario, i.e., U of Waterloo, Queen's, Ryerson karate clubs.

Psychology Graduate Students Association, Department of Psychology, Faculty of Arts & Science,

University of Toronto

Toronto, ON

Academic Poster Judge

May 2026

- Evaluated four academic poster presentations by graduate students enrolled with the Department of Psychology based on criteria of academic rigour, delivery, and presentation style established by the Chair of an academic psychology conference.

Shorinji-Kan Jiu Jitsu Club, Fitness and Wellness, Hart House, University of Toronto

Toronto, ON

Assistant Instructor

October 2025 — Present

- Assist with the instruction of beginner and intermediate classes.
- Demonstrate core techniques and supervise technical drills involving takedowns, throws, joint manipulations, and break falling.
- Provide helpful feedback to support skill acquisition with an emphasis on control and safety.
- Maintain a safe and supportive learning environment by assisting with first aid in cases of injury.
- Help manage warm-ups, class structure, training flow, and cooldowns with the Head Instructor.

University of Toronto Karate Club, Club Sports, Sport & Rec, Faculty of Kinesiology & Physical

Education (KPE), University of Toronto

Toronto, ON

Vice President

June 2025 — Present

- Organize various events such as the Inter-University Karate tournament for karate students across Ontario, providing over 100 athletes with a post-secondary competitive environment
- Lead various social activities within the club and managing the team's online presence
- Manage member status of current athletes, including fees, training, grading, attendance and relations with external institutions such as the Ontario Karate Federation and other clubs across Ontario, i.e., U of Waterloo, Queen's, Ryerson karate clubs

University of Toronto Psychedelics Journal Club

Toronto, ON

Discussion Leader & Presenter

December 2024 — April 2025

- Led discussions on recent publications in the psychedelics literature on the utility of psychedelic substances in therapy and for the exploration of altered states of consciousness as they pertain to frameworks in cognitive science.
- Presented notable findings (of choice) in the literature, offering a wide-ranging scope of the theoretical basis of such findings and their implications for the empirical and practical uses of psychedelic substances, i.e., in research and in the clinic.

Freelance Academic Tutoring Services

Hong Kong, China (Remote) & Toronto, ON

Highschool English & Science Tutor

May 2022 — January 2025

- Taught struggling students senior-level high school courses in English, Calculus & Biology.
- Created additional resources in the form of lectures, worksheets, annotations, citation guides, writing guides, essay assignments, and mock examinations related to the Ontario Secondary School Literacy Test (OSSLT) and the IELTS.

CERTIFICATIONS

WHMIS Training, University Health Network	2026
Accessibility for Ontarians with Disabilities Act (AODA) Training, Baycrest Health Sciences	2026
Safe Sport Training, Coaching Association of Canada	2025
TCPS2 CORE 2022	2025
MathWorks Core MATLAB Skills, Computer Vision Onramp, & Deep Learning Onramp	2025
Certificate of Completion, Montreal Cognitive Assessment (MoCA)	2025
CIHR Institute of Gender and Health Core Competency Modules for Sex and Gender in Biomedical Research, Primary Data Collection with Human Participants, and the Analysis of Secondary Data from Human Participants	2025
The Johns Hopkins University, Fundamental Neuroscience for Neuroimaging Course	2024
The University of Hong Kong, Academy for the Talented, Junior Neuroscience Program	2021

RELEVANT COURSEWORK

PSY359 Human Neuroimaging Laboratory (A+)	PSY380 Vision Science (A)
PSY312 Cognitive Development (A+)	PSY396 Neurochemical Basis of Behaviour (A)
PSY309 Research Specialization: Practicum (A+)	PSY290 Behavioural Neuroscience (A+)
PSY405 Independent Research Project (A+)	COG341 Attention, Perception, Consciousness (A)

TECHNICAL SKILLS

Programming & Data Analysis	R, Python, MATLAB, JavaScript, HTML5/CSS, jamovi, JASP, NVivo (qualitative analysis), Taguette
Neuroimaging Software	FSL (FEAT/FLIRT/BET/MELODIC/TBSS/FSLEYES)
Experiment Design & Programming	PsychoPy, Psychtoolbox (MATLAB), Qualtrics, Lookit, Pavlovia, REDCAP
Technical Equipment	EyeLink 1000 Plus, EyeLink DataViewer, EDF2ASC
Databases & Literature Review Tools	Zotero, Covidence, PubMed, PsycINFO, JSTOR, Scopus, Google Scholar, Mendeley, Excel/Notion
Design & Graphics	Adobe Photoshop, Adobe Lightroom, Figma, Canva, Blender, Material Design
User App Testing	TestFlight (user testing as an UX researcher)

LANGUAGES

Native	English, Mandarin
Fluent	Heritage Cantonese, Heritage Taiwanese Hokkien
Conversational	French, Japanese